Task: Exploring Tokenization Across Models and Languages

Objective:

Understand how different tokenization algorithms work in LLMs by calculating token counts for various texts across different models and languages. This will help you grasp the importance of tokenization in model efficiency and cost.

Instructions:

Calculate a number of tokens required for each version of text for GPT-3.5-Turbo, GPT-4, and GPT-4o models. (Optionally) compare with Mistral Nemo and LLAMA 3.1 tokenizers.

Hint: Use tokenizer libraries to calculate the number of tokens.

Texts:

Case 1: English:

```
The cat sat on the windowsill, watching the rain fall. Suddenly, a flash of lightning
lit up the sky, startling the little creature. It leaped down and scurried to its
favorite hiding spot under the bed.
```

Case 2: Spanish:

```
El gato estaba sentado en el alféizar de la ventana, mirando la lluvia caer. De
repente, un relámpago iluminó el cielo, sobresaltando a la pequeña criatura. Saltó y
corrió a su escondite favorito debajo de la cama.
```

Case 3: Arabic:

```
جلست القطة على حافة النافذة، تراقب هطول المطر. وفجأة، أضاءت ومضة من البرق السماء، مما
أثار ذهول المخلوق الصغير. قفزت إلى أسفل وهرعت إلى مكان اختبائها المفضل تحت السرير.
```

Case 3: Hindi :

```
बिल्ली खिड़की पर बैठी हुई बारिश को देख रही थी। अचानक, आसमान में बिजली चमकी, जिससे छोटा जीव चौंक गया। वह
नीचे कूद गई और बिस्तर के नीचे अपनी पसंदीदा छिपने की जगह पर भाग गई।
```

Expected results:

- Provide table with token counts for each model and language.
- Explain in free from your observations and reflections on the tokenization process and its implications for LLMs.

Assessment Criteria:

- **Accuracy:** Correct token counts and understanding of tokenization principles.
- **Reflection:** Insightfulness of discussion and reflection.